Run report: lit2026 W12 zero-minor scaling, run-20261012T0009Z

Descriptive run report (observations only). The interpretation is in the study's `README.md` and `REPORT.pdf`.

What ran

- **Binary.** `examples/lit2026_w12_zero_minor.rs`, built in release mode.
  - The scaling run and its replay used commit `c70d57ec7`; binary sha256 `02aaf7cb…2740`.
  - The defect sweep, `refit` and `compare` used commit `06b032f0c`. That commit changes only the fit guard and adds two analysis modes; the solver code is identical.
  - The rerun d-sweep gave the same per-run counts as the sweep that crashed after its last run (`dsweep-failed-c70d57ec7.log`, `diff` clean).
- **Host.** Shared M4 Pro (macOS, Darwin 25.6.0), isolation **L0**. `uptime` was recorded with every sample in `scaling.json` (load averages 79–106 at the start, 97–104 at the end). Wall time is reported only as L0 medians. The gate is counted operations.
- **Commands** (run inside this directory):
  - `lit2026_w12_zero_minor run --bits 16,18,20,22,24,26 --targets 8 --rho-seeds 4 --out scaling.json`
  - `lit2026_w12_zero_minor dsweep --bits 20 --ds 2,3,4 --targets 8 --out dsweep.json`
  - `lit2026_w12_zero_minor refit --in scaling.json --out fits.json`
  - The run repeated once more, then `compare --a scaling.json --b <replay>.json` (`replay-compare.txt`).
- **Curves.** The registered prime-order ladder `find_prime_order_curve(bits, 59297)`, cofactor 1. The binary rebuilds each curve from its registry parameters and asserts the ICV1 slug:
  - `icv1-fp16-t295-8d3c3165`, `icv1-fp18-tm337-d28d5e09`, `icv1-fp20-t727-cd198a38`;
  - `icv1-fp22-tm1385-475dcb5f`, `icv1-fp24-t1059-6df599df`, `icv1-fp26-tm1775-7e8fb6df`.
  - The generator is the point with the least abscissa and the smaller ordinate.
- **Targets.** 8 per curve, planted `k` from seed 20261011. The solvers never see `k`. Each answer is verified outside the solver by `[m]G = Q`.
- **Arms, interleaved target by target.**
  - `zero_minor.guess_determine`: 1 seed per target.
  - `rho.negation` (`ic_boundary::rho_walk_with` + `NegationClasses` + `RhoWalk::negation()`, cap 64×floor): 4 walk seeds per target.
- **Unit.** F_p multiplications (squarings included) plus inversions, each counted once. Additions are free; this is the convention of `ic_boundary::FieldOps`. Rho's ops come from `PrimeCurve`'s counting group law. The zero-minor ops come from the example's counted `Fp`, plus `PrimeCurve`'s count for the 2ℓ scalar multiplications.

Parameters chosen by the paper's rules

bits	r	n'	$\ell = 3n'$	ℓ'	d (Eq. 4 ≥ 0.9)	Eq. (4) success	distinct maximal minors per guess
16	53117	14	42	21	3	1.0000	1.81e6
18	212437	16	48	24	3	1.0000	4.17e6
20	562427	18	54	27	3	1.0000	8.68e6
22	2435809	20	60	30	3	0.9999	1.67e7
24	16482251	22	66	33	3	0.9059	3.00e7
26	42653927	24	72	36	4	1.0000	3.52e9

Results: every run solved and verified (48/48 zero-minor, 192/192 rho)

bits	rho.negation median ops (min–max), 32 runs	zero-minor median ops (min–max), 8 runs	zero-minor search-phase median	paired ratio zm/rho, geomean (95% bootstrap)	zm median wall ms (L0)	rho median wall ms (L0)
16	1.92e3 (1.14e3–6.30e3)	5.53e5 (3.14e5–8.30e5)	2.22e5	260 (215–319)	9.0	0.067
18	2.98e3 (1.43e3–5.13e3)	1.48e6 (6.74e5–2.74e6)	9.43e5	455 (326–626)	75	0.11
20	4.62e3 (1.59e3–8.64e3)	2.54e6 (9.15e5–6.30e6)	1.71e6	563 (350–883)	262	0.19
22	7.93e3 (3.12e3–2.05e4)	9.03e6 (1.42e6–2.13e7)	7.79e6	785 (349–1633)	1660	0.44
24	1.67e4 (4.51e3–4.61e4)	2.16e7 (6.92e6–5.82e7)	1.98e7	1109 (706–1743)	4776	0.66
26	2.49e4 (6.48e3–6.29e4)	2.71e8 (7.76e6–5.99e8)	2.68e8	6715 (2263–15336)	51628	1.12

Fits of $\log_2$ cost against $\log_2 r$ (`fits.json`; OLS over runs, 95% bootstrap resampling runs within each curve, t interval in brackets after it):

series	slope	95% bootstrap	95% t
rho.negation, field ops (n = 192)	0.377	0.350–0.404	0.350–0.404
rho.negation, group ops (n = 192)	0.400	0.371–0.430	0.371–0.430
zero-minor, field ops, all phases (n = 48)	0.783	0.669–0.879	0.669–0.897
zero-minor, field ops, search phases only (n = 47)	0.905	0.769–1.021	0.746–1.064
zero-minor, d-subsets covered (n = 47)	0.929	0.793–1.048	0.763–1.095
zero-minor, d = 3 runs only, 16–24 bits (n = 40)	0.631	0.542–0.717	0.528–0.734

The "search phases only" and "covered" fits have n = 47. One 16-bit target was solved by a zero entry of the dense part, before any determine step.

Covered d-subsets divided by r, median per curve: 1.37, 1.32, 0.92, 0.99, 0.37, 2.13 (16 ⟶ 26 bits); the ranges are in `fits.json`.

The run's own fits in `scaling.json` group by method string, and that string carries d. Its `d=4` entries rest on a single curve size and are degenerate; use `fits.json` instead.

Defect sweep on `icv1-fp20-t727-cd198a38` (8 targets per d, same targets). Median ops (min–max):

- $d = 2$: 1.21e7 (9.58e5–4.25e7)
- $d = 3$: 2.54e6 (9.15e5–6.30e6)
- $d = 4$: 3.32e6 (9.11e5–9.88e6)

All 24 runs verified.

Replay. A second full execution of the scaling run recorded identical method, seed, op counts, group ops, search ops, verification and recovered value in 240 of 240 rows (`replay-compare.txt`).

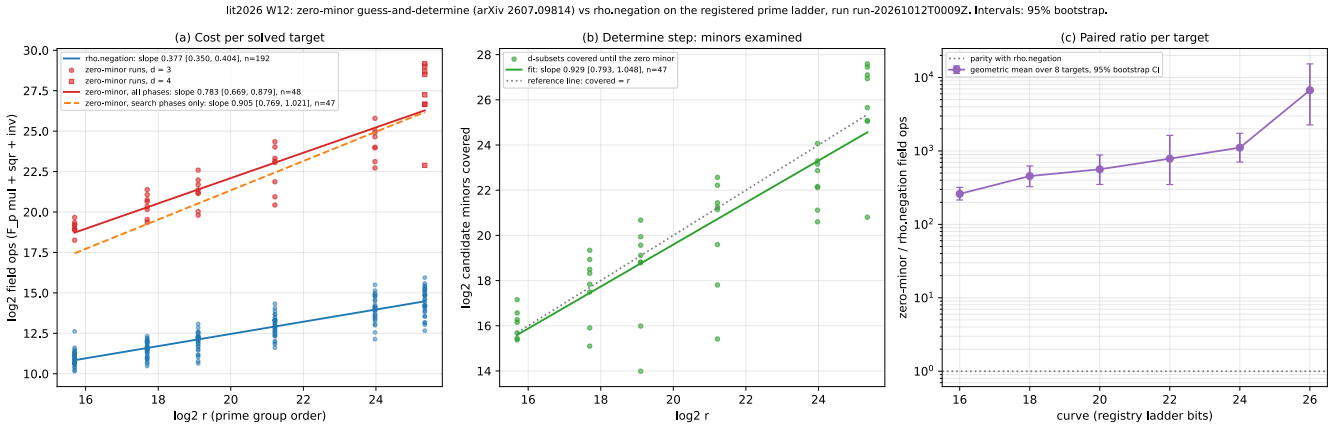


Figure 1: Run-20261012T0009Z. (a) Field operations per solved target, against $\log_2 r$: rho.negation (32 runs per curve) and zero-minor (8 runs per curve; circles $d = 3$, squares $d = 4$), with the OLS fits from `fits.json`. (b) d-subsets of the signature matrix covered before the zero minor, with the line covered $= r$ for reference. (c) Paired per-target ratio of zero-minor to rho field ops, geometric mean with 95% bootstrap interval.

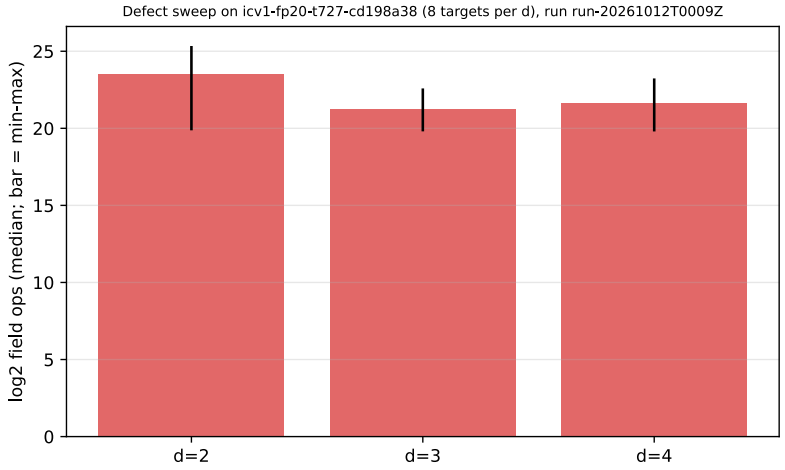


Figure 2: Defect sweep, run-20261012T0009Z, 8 targets per d on the 20-bit ladder curve: median field ops, with the bar spanning min to max.

Raw records

`scaling.json` (every run, with per-phase tallies, counters and `uptime`), `scaling.log`, `fits.json`, `dsweep.json`, `dsweep.log`, `dsweep-failed-c70d57ec7.log` (a crash in the fit step after all runs: a single-size fit gave NaN), `replay-compare.txt`, `uptime-start.txt`, `uptime-end.txt`. The plot source is `../../plot_scaling.py`; it computes nothing.